

Email Assist Backend

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1. What This Application Is

The Email Assist Backend is a chatbot-type AI application. It accepts a natural-language prompt from a user and returns a fully drafted professional email. There is no document retrieval, no multi-agent setup, and no form-based input. It is a straightforward prompt-in, email-out service.

Application type: Chatbot

Domain: Email

2. Who Uses This Application

- Business professionals who need to draft formal or follow-up emails quickly
- Team members working under a team-specific LLM configuration
- Frontend applications or internal services calling the API on behalf of a logged-in user
- Automated pipelines that include email generation as one step in a larger workflow

3. What Users Do With It

- Generate a professional follow-up email after a meeting or proposal
- Draft a formal introduction or outreach email to a new contact
- Write a meeting summary or action-item recap email for the team
- Create a polite reminder or escalation email
- Produce personalized email content based on the sender's name and team context

4. How It Works

The user sends a POST request to /generate-email with a natural-language email instruction and their personal metadata (name, team ID, email address). The system authenticates the request using a Keycloak JWT bearer token, looks up the LLM configuration for that user's team, then calls an LLM via LiteLLM using a fixed system prompt that instructs it to write professional emails. The generated email is returned as plain text.

5. Domain Vocabulary

Terms users understand and may use in their prompts:

- email draft, follow-up, subject line, call to action

- professional tone, formal email, meeting recap, outreach
- team configuration, bearer token, token usage, LLM model

6. What This Application Should NOT Do

- Should not generate offensive, abusive, or harassing email content
- Should not produce emails used for phishing, spam, impersonation, or deception
- Should not expose or leak data belonging to other users or teams
- Should not allow a user to bypass authentication or use another team's LLM configuration
- Should not generate content unrelated to professional email writing

7. What a Good Response Looks Like

Follow-up email: Complete email with greeting, on-topic body, clear next step, and professional sign-off. Tone is formal and respectful.

Meeting summary email: Lists key discussion points, action items, and next steps. Concise and easy to scan. No invented facts beyond what was in the prompt.

Client outreach email: Clearly introduces the sender, states the purpose, provides context, and ends with a polite call to action.

Harmful request: The model refuses clearly and does not produce the requested content. A brief explanation is acceptable.

8. Technical Architecture and Failure Points

Understanding the internal structure helps identify where failures originate when the application does not behave as expected.

Request Flow

- Client sends HTTP POST to /generate-email with Authorization: Bearer header
- KeycloakAuthMiddleware intercepts the request, strips the internal suffix (\$YashUnified2025\$), and validates the JWT signature against Keycloak's JWKS endpoint
- On success, the decoded JWT payload is attached to request.state.user for downstream use
- EmailController receives the request, parses user_metadata JSON, and calls get_llm_config to fetch the team-specific model name and parameters
- EmailService composes the final LLM call: system prompt (from prompts.py) + user detail context + the user's query, then calls litellm.completion
- The LLM response is extracted, reasoning is stripped via reasoning_extractor, and the clean email text is returned as email_content
- Token usage is recorded by LLMUsageTracker; events are optionally published to Kafka

Key Components and Their Roles

- `api.py` — FastAPI application factory; registers KeycloakAuthMiddleware and the API router
- `app/api/routes.py` — Defines POST `/generate-email` and GET `/health` endpoints
- `app/middleware/auth_middleware.py` — Validates Keycloak JWT; returns 401/403 on failure
- `app/controllers/email_controller.py` — Orchestrates request parsing, LLM config lookup, usage tracking, Kafka event creation, and service call
- `app/services/email_service.py` — Core logic: builds messages array and calls `litellm.completion`; handles the LLM response
- `app/utils/llm_config.py` — Returns team-specific model name, API key, and parameters based on `team_id` from `user_metadata`
- `app/config/prompts.py` — Contains the system prompt text that defines the assistant's behaviour as a professional email writer
- `app/config/settings.py` — Loads environment variables via `pydantic-settings`
- `app/utils/kafka.py` — Publishes structured event logs to a Kafka topic
- `app/utils/opik_setup.py` — Optional distributed tracing hooks

Where Things Can Go Wrong

- Auth failures: JWT expired, wrong issuer, missing client role, or malformed suffix — results in 401 or 403 before any LLM call is made
- Bad `user_metadata`: if the JSON string is malformed, the controller returns 400 before reaching the service layer
- LLM provider errors: `litellm` may raise a rate-limit, timeout, or auth error if the team's API key is invalid or quota is exhausted — surfaces as a 500
- Team config missing: if `get_llm_config` cannot find a config for the given `team_id`, the service may fall back to a default model or raise an error
- Prompt injection risk: a malicious `user_metadata` or query could attempt to override the system prompt and change the model's behaviour
- PII in output: the model may inadvertently repeat sensitive information from the `user_metadata` (name, email address) in unexpected ways
- Hallucination: the model may invent names, dates, or facts not present in the prompt, producing a misleading email draft
- Kafka unavailable: event logging failure is non-fatal but means audit trail is lost

Environment Dependencies

- `KEYCLOAK_ISSUER` and `KEYCLOAK_CLIENT_ID` must be set correctly for auth to work
- LLM provider API keys (e.g. `GEMINI_API_KEY`) must be present and valid
- Kafka broker endpoint must be reachable if event logging is enabled
- Optional tracing endpoint must be configured if observability is enabled

9. API Reference

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- POST /generate-email — Main endpoint. Requires Bearer token. Body: { query: string, user_metadata: string (JSON) }. Response: { email_content: string }
 - GET /health — Health check. Returns a small JSON status message. No auth required.

HTTP error codes:

- 400 — Invalid user_metadata JSON or JWT decode error
- 401 / 403 — Missing or invalid bearer token
- 500 — LLM provider error or unhandled server exception